

S&S QUIZ-4 SOLUTION

Sol 1(a)÷ Given -

$$y(t) = \int_{-\infty}^t e^{-(t-\tau)} \cdot x(\tau-3) d\tau$$

Let $\tau-3 = \lambda$

$$d\tau = d\lambda$$

$$y(t) = \int_{-\infty}^{t-3} e^{-(t-\lambda-3)} \cdot x(\lambda) d\lambda \rightarrow (1 \text{ Mark})$$

$$= \int_{-\infty}^{\infty} e^{-(t-\lambda-3)} \cdot u(t-\lambda-3) \cdot x(\lambda) d\lambda \quad \text{--- (1)} \\ \rightarrow (1 \text{ Mark})$$

As we know that - $y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\lambda \quad \text{--- (2)}$

$\rightarrow (1 \text{ Mark})$

Comparing eqⁿ (1) & eqⁿ (2), we have -

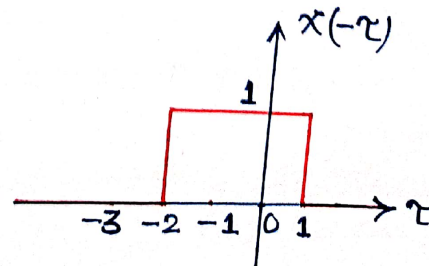
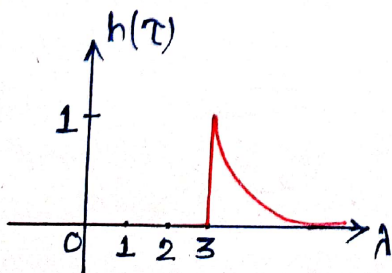
Impulse response, $h(t) = e^{-(t-3)} \cdot u(t-3) \rightarrow (2 \text{ Mark})$

Sol 1(b)÷ Given - $x(t) = u(t+1) - u(t-2)$

$$h(t) = e^{-(t-3)} \cdot u(t-3)$$

Impulse Response of the system, $y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\lambda$

$$= \int_{-\infty}^{\infty} x(t-\tau) \cdot h(\tau) d\lambda \rightarrow (1 \text{ Mark})$$



$\rightarrow (2 \times 0.5 \text{ Mark})$

Case(1)÷ For $t \leq 2$, $y(t) = 0$

→ (1 Mark)

Case(2)÷ For $2 < t \leq 5$, $y(t) = \int_3^{t+1} 0 \cdot e^{-(\tau-3)} d\tau$

$$= \left\{ \frac{e^{-(\tau-3)}}{-1} \right\}_3^{t+1}$$

$$= - \left\{ e^{-(t+1-3)} - e^{-(3-3)} \right\}$$

$$= 1 - e^{-(t-2)}$$

→ (1 Mark)

Case(3)÷ For $5 < t$, $y(t) = \int_{t-2}^{t+1} e^{-(\tau-3)} d\tau$

$$= \left\{ \frac{e^{-(\tau-3)}}{-1} \right\}_{t-2}^{t+1}$$

$$= - \left\{ e^{-(t+1-3)} - e^{-(t-2-3)} \right\}$$

$$= e^{-(t-5)} - e^{-(t-2)}$$

→ (1 Mark)